NADER ABDULMALEK AL-GHAFRI

Biomedical Engineer

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PROFILE

Biomedical engineer with 17 months of hospital maintenance work on sterilization, hemodialysis and anesthesia equipment. Calibrates devices, replaces internal components and traces faults from the alarm to the failed part.

EXPERIENCE

Anesthesia Equipment Maintenance 8 months

- Ran preventive and corrective maintenance on anesthesia workstations: ventilator modules, vaporizers, gas flow systems and breathing circuits.
- Calibrated flow and pressure sensors and performed leak tests before releasing machines to the operating room.

Sterilization Equipment Maintenance 6 months

Al-Mouwasat University Hospital, Damascus

- Maintained steam autoclaves and sterilization units, including door seals, valves, heaters and control boards.

Hemodialysis Equipment Maintenance 3 months

- Serviced hemodialysis machines: blood pumps, hydraulic circuits, and conductivity and temperature sensors.

EDUCATION

B.Sc. Biomedical Engineering

Al-Andalus University for Medical Sciences, Syria

PROJECTS

Polymer Nanostructure Formation for Theranostics

Graduation project

- Designed a polymer nanostructure that combines diagnosis and targeted therapy in one carrier.

Fluorescence-Based Detection of Cancer Cells

Applied project

- Developed a fluorescence-based method to detect cancer cells.

RESEARCH

- Presented a paper on fluorescence-based cancer cell detection at a biomedical engineering conference, 2025.

TRAINING

- Echocardiography and Diagnostic Ultrasound: operating ultrasound systems, probe selection and image acquisition.

SKILLS

- **Equipment maintenance:** calibration, fault diagnosis, component replacement, preventive and corrective maintenance
- **Engineering:** medical device design, biomaterials, biomechanics
- **Software:** MATLAB, Python, SQL (data analysis), Proteus 8 Professional
- **Professional:** teamwork, problem solving, time management, Agile methods, work under pressure

LANGUAGES

Arabic: native | English: intermediate (B1)